

Sinusoidal Waves

Wavelength, periodic time, frequency and the wave equation $v = f\lambda$, for transverse and longitudinal waves.

Grade 11 · Egyptian Baccalaureate

14 questions

Step-by-step solutions

 $v = f\lambda$

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PHYSICS

The toolkit for this lesson

$$v = f\lambda$$

the wave equation

$$v = \lambda / T$$

the same thing, written with T

$$f = 1 / T$$

frequency and periodic time

y – x graphgives the amplitude A and the wavelength λ **y – t graph**

gives the amplitude A and the periodic time T

transverse / longitudinalparticles $\perp$ / $\parallel$ to the travel direction

A wave carries energy, not matter. Every particle of the medium simply vibrates about its own fixed position; nothing travels along with the wave except the disturbance and the energy it carries. And one equation, $v = f\lambda$, answers almost every numerical question in this lesson — the only skill is deciding which two of the three quantities you have been given.

CLASSWORK 1

A

Frequency, wavelength and speed

- 1 A vibrating body completes one full oscillation in a periodic time of 0.20 s. **Find its frequency.**

PROBLEM

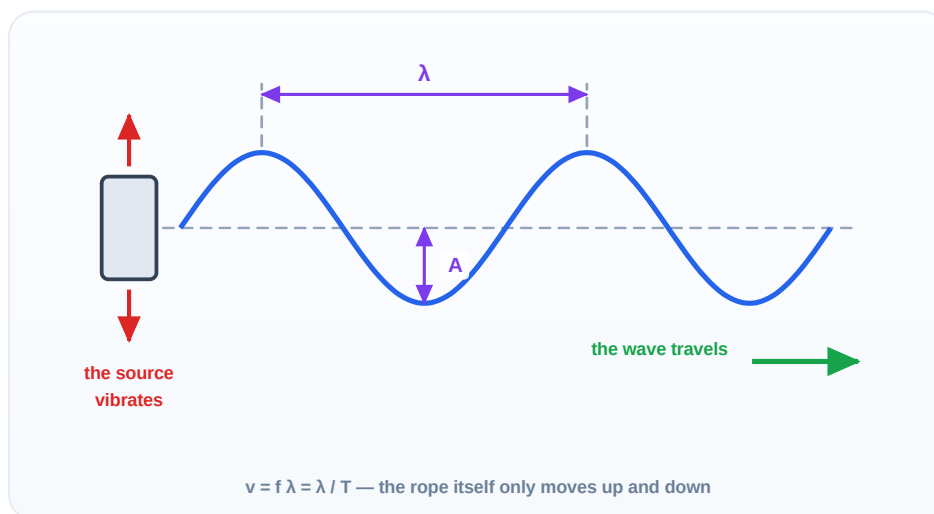


FIGURE 1

- 2 A sinusoidal wave has a wavelength of 0.2 m and a frequency of 0.3 Hz. **Calculate the speed with which the wave travels.**

PROBLEM

What a wave carries

3 Which of the following is transported by a mechanical wave as it travels through the medium?

MCQ

A matter

B energy

C the particles of the medium

D the total displacement

4 A wave has a periodic time equal to 0.25 second. What is its frequency?

MCQ

A 0.25 hertz

B 2.5 hertz

C 4.0 hertz

D 0.4 hertz

5 A wave travels along a string; its periodic time is 0.10 s and its wavelength is 1.5 m.
Calculate the speed of the wave.

PROBLEM

6 A wave pulse travels along a rope with a speed of 4.0 m/s and a wavelength of 0.80 m.
Find its frequency and its periodic time.

PROBLEM

7 The opposite figure shows the vibration of the particles of a material medium for a longitudinal wave motion. **Using the scale shown in the figure, calculate the wavelength of the longitudinal wave.**

PROBLEM

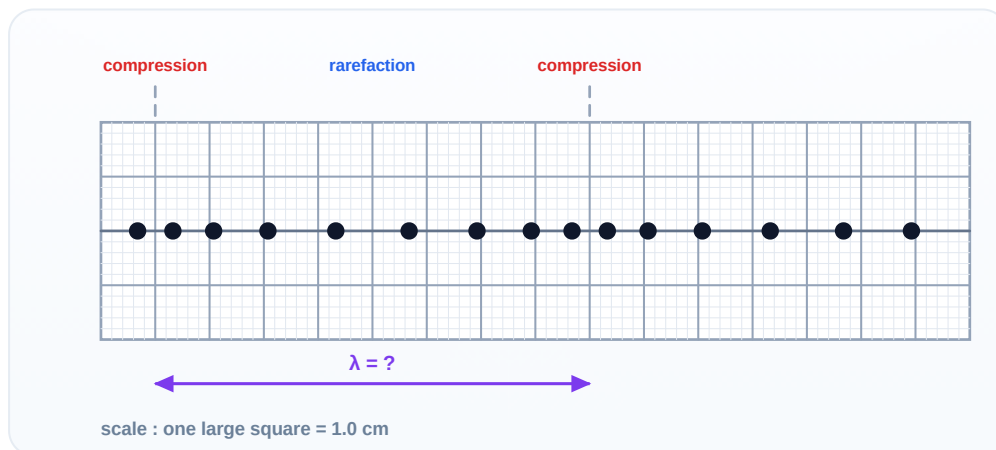


FIGURE 2

Using the wave equation

- 8 A sinusoidal wave has a wavelength of 0.60 m, and its periodic time equals 0.20 s.
Calculate:

PROBLEM

- a) the frequency
b) the speed of the wave

- 9 A wave source vibrates with a periodic time of 0.25 s, and produces a wave whose wavelength is 1.2 m. **Calculate the speed of the wave.**

PROBLEM

Reading the two graphs

- 10 Which of the graphs is used to determine **both** the amplitude and the wavelength of a wave motion made by a vibrating body, directly?

MCQ

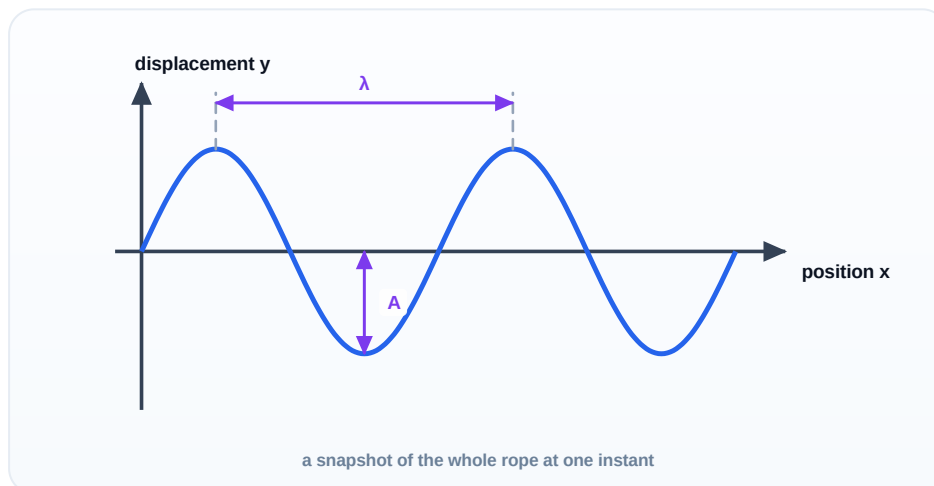


FIGURE 3

- A the displacement – time graph ($y - t$)
B the displacement – position graph ($y - x$)
C the velocity – time graph
D the frequency – periodic time graph

11 In a longitudinal wave, the particles of the medium vibrate:

MCQ

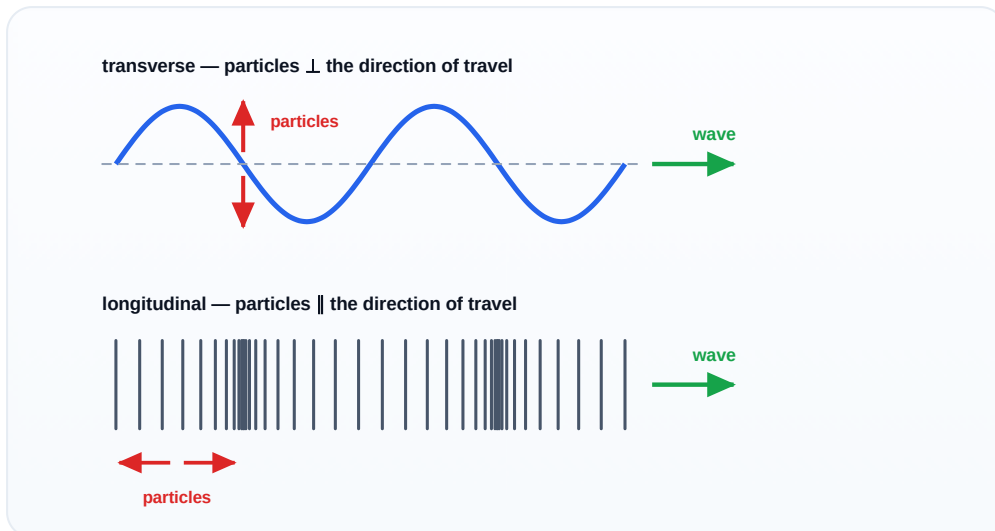


FIGURE 4

A perpendicular to the direction of propagation of the wave

B in a direction parallel to the direction of propagation of the wave

C in circular paths opposite to the direction of transfer of the energy

D with a constant speed equal to the speed of light

12 A sound wave travels through air with a speed of 340 m/s and a frequency of 500 Hz. Calculate its wavelength.

PROBLEM

13 The displacement – position graph of a sinusoidal wave shows a wavelength of 3.0 m and an amplitude of 0.50 m, while the displacement – time graph shows a periodic time of 0.60 s. Find the speed of the wave.

PROBLEM

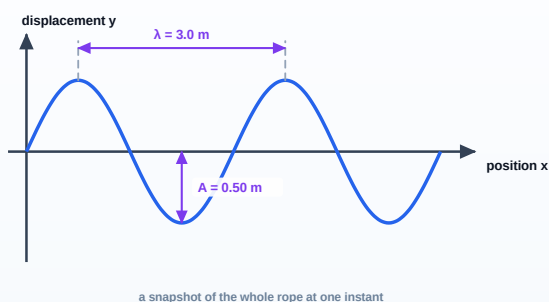


FIGURE 5

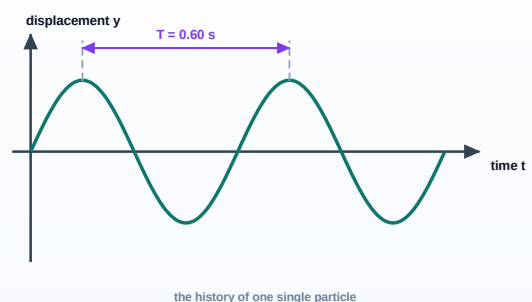


FIGURE 6

- 14 The opposite figure represents a sinusoidal wave motion. If the distance between the two points P and Q is 2 m, covered in a time interval of 0.1 s, **calculate:**

- a) the wavelength
- b) the speed of propagation of the wave

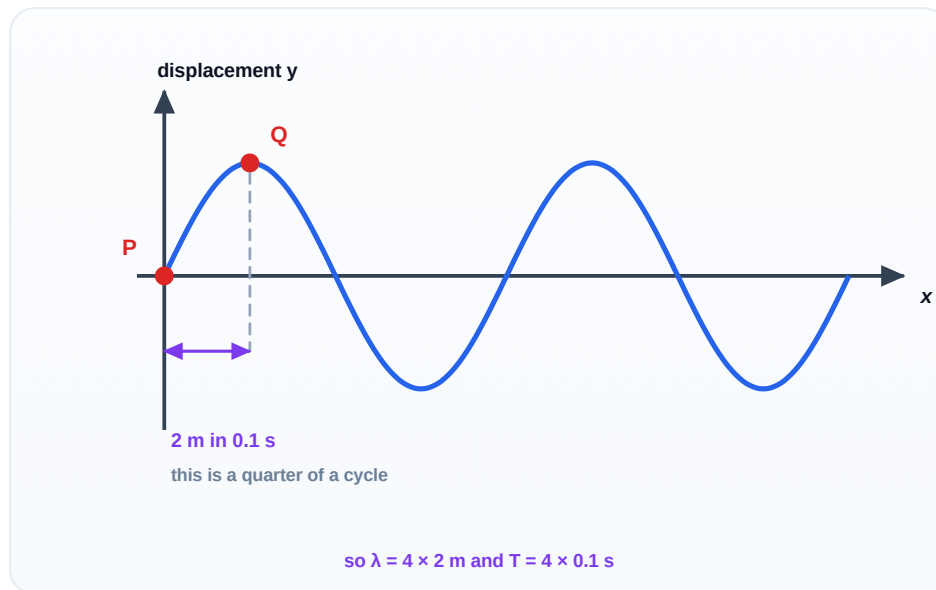


FIGURE 7



Solutions — step by step

One equation, three faces. $v = f\lambda$, $f = 1/T$ and $v = \lambda/T$ are the same statement written three ways. Read the question, write down which two quantities you were handed, and the third one follows in a single line.

Part A — Classwork 1

- 1 the frequency for a periodic time of 0.20 s

$$f = 1 / T$$

$$f = 1 / 0.20$$

$$f = 5 \text{ Hz}$$

The frequency is simply how many complete oscillations happen each second. One oscillation every fifth of a second means five of them every second.

$$f = 5 \text{ Hz}$$

- 2 the speed of a wave with $\lambda = 0.2 \text{ m}$ and $f = 0.3 \text{ Hz}$

$$v = f \lambda$$

$$v = 0.3 \times 0.2$$

$$v = 0.06 \text{ m/s}$$

Each second the source sends out 0.3 of a wavelength, and each wavelength is 0.2 m long, so the front of the wave advances 0.06 m in that second. That is what $v = f\lambda$ is saying.

$$v = 0.06 \text{ m/s}$$

Part B — Home assignment 1

- 3 what a mechanical wave transports

each particle of the medium vibrates about its own fixed position and stays there.
what moves forward is the disturbance, and with it the **energy** of the vibration.

A cork floating on a ripple bobs up and down, it does not travel across the pond. The matter stays put; only the energy is passed on from one particle to the next.

B) energy

- 4 the frequency for $T = 0.25 \text{ s}$

$$f = 1 / T = 1 / 0.25 = 4.0 \text{ Hz}$$

C) 4.0 hertz

5 a wave on a string, $T = 0.10$ s and $\lambda = 1.5$ m

$$v = \lambda / T$$

$$v = 1.5 / 0.10$$

$$v = 15 \text{ m/s}$$

You could also find $f = 1/0.10 = 10$ Hz first, then $v = f\lambda = 10 \times 1.5 = 15$ m/s. The two routes are the same equation.

$$v = 15 \text{ m/s}$$

6 a pulse of $v = 4.0$ m/s and $\lambda = 0.80$ m

$$v = f \lambda \quad \rightarrow \quad f = v / \lambda$$

$$f = 4.0 / 0.80 = 5 \text{ Hz}$$

$$T = 1 / f = 1 / 5$$

$$T = 0.20 \text{ s}$$

$$f = 5 \text{ Hz}, T = 0.20 \text{ s}$$

7 the wavelength of the longitudinal wave from the figure

in a longitudinal wave the wavelength is the distance between the centres of two **successive compressions** (or two successive rarefactions).

from the figure, the two compression centres are 8 large squares apart :

$$\lambda = 8 \times 1.0 \text{ cm}$$

$$\lambda = 8.0 \text{ cm} = 0.080 \text{ m}$$

Note on the printed figure. The figure in the book is drawn on a squared grid but no numerical scale is printed beside it, so the question cannot be answered from the book as it stands. The figure has been redrawn here with the scale stated explicitly (one large square = 1.0 cm). The physics is unchanged: find two successive compressions, count the squares between their centres, and multiply by the scale. Measuring between a compression and the **next rarefaction** would give only half a wavelength — that is the usual mistake.

$$\lambda = 8.0 \text{ cm} = 0.080 \text{ m}$$

Part C — Classwork 2

8 $\lambda = 0.60$ m and $T = 0.20$ s

a) $f = 1 / T$

$$f = 1 / 0.20 = 5 \text{ Hz}$$

b) $v = f \lambda$

$$v = 5 \times 0.60$$

$$v = 3.0 \text{ m/s}$$

Check it the short way: $v = \lambda/T = 0.60/0.20 = 3.0$ m/s, the same answer without finding f at all.

$$\text{a) } f = 5 \text{ Hz} \quad \text{b) } v = 3.0 \text{ m/s}$$

9 a source of $T = 0.25$ s producing $\lambda = 1.2$ m

$$v = \lambda / T$$

$$v = 1.2 / 0.25$$

$$v = 4.8 \text{ m/s}$$

Notice that the source decides the **frequency** and the medium decides the **speed**; the wavelength is then whatever v/f happens to be.

$$v = 4.8 \text{ m/s}$$

Part D — Home assignment 2

10 which graph gives A and λ

the $y - x$ graph is a **snapshot** of the whole medium at one instant, so :

the height of a crest above the axis $\rightarrow$ the amplitude A

the distance between two crests $\rightarrow$ the wavelength λ

the $y - t$ graph follows one single particle, so it gives A and T , not λ .

The two graphs look identical, and that is exactly the trap. Always read the **horizontal axis** first: if it is a distance, the repeat length is λ ; if it is a time, the repeat length is T .

B) the displacement – position graph ($y - x$)

11 the vibration of the particles in a longitudinal wave

longitudinal $\rightarrow$ each particle vibrates **along** the same line on which the wave travels, producing compressions and rarefactions (sound is the example).

transverse $\rightarrow$ each particle vibrates **across** that line (a rope, light).

B) in a direction parallel to the direction of propagation of the wave

12 the wavelength of a 500 Hz sound wave

$$v = f \lambda \quad \rightarrow \quad \lambda = v / f$$

$$\lambda = 340 / 500$$

$$\lambda = 0.68 \text{ m}$$

This is about the length of your forearm — and it is why a 500 Hz sound bends easily round a doorway, while light, with a wavelength a million times smaller, does not.

$$\lambda = 0.68 \text{ m}$$

13 reading the speed from the two graphs

the $y - x$ graph gives $\lambda = 3.0$ m (and $A = 0.50$ m)

the $y - t$ graph gives $T = 0.60$ s

$$v = \lambda / T$$

$$v = 3.0 / 0.60$$

$$v = 5.0 \text{ m/s}$$

The amplitude 0.50 m is not needed for the speed — it is there to see whether you know that the amplitude never enters $v = f\lambda$.

$$v = 5.0 \text{ m/s}$$

14 P and Q a quarter of a cycle apart

from the figure, P is at the equilibrium level and Q is at the next crest, so PQ is a **quarter** of a complete wave :

a) $\lambda / 4 = 2 \text{ m}$
 $\lambda = 4 \times 2 = 8 \text{ m}$

b) $T / 4 = 0.1 \text{ s} \quad \rightarrow \quad T = 0.4 \text{ s}$
 $v = \lambda / T = 8 / 0.4$
 $v = 20 \text{ m/s}$

The quarter-cycle is the whole question. From the equilibrium level to the next crest is a quarter of a wave, from crest to the next crest is a whole wave, and from crest to trough is half a wave. Get that right and both parts follow at once. As a check, $f = 1/0.4 = 2.5 \text{ Hz}$ and $v = f\lambda = 2.5 \times 8 = 20 \text{ m/s}$.

a) $\lambda = 8 \text{ m}$ b) $v = 20 \text{ m/s}$